Chapter 10: Application Layer



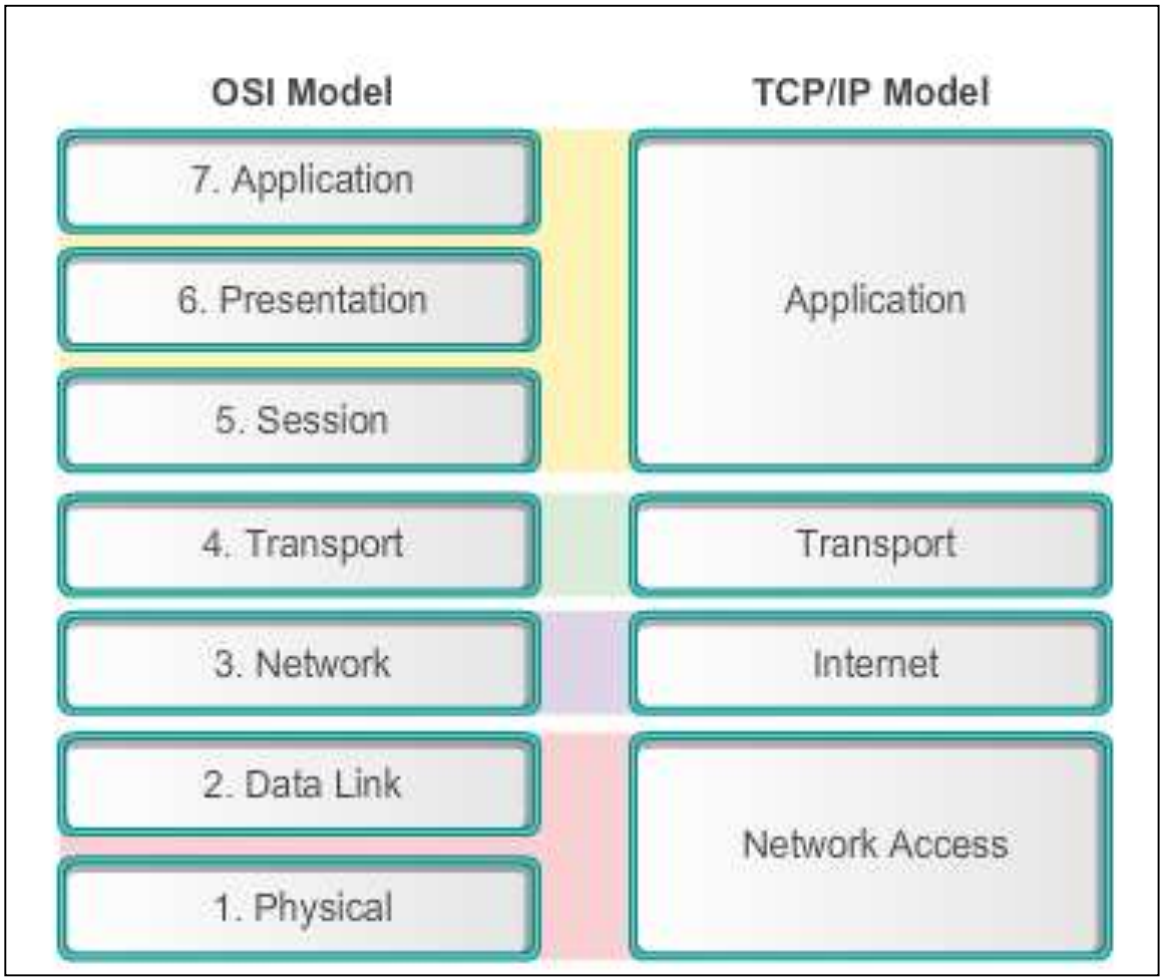
Introduction to Networks

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Application, Session and Presentation

OSI and TCP/IP Models Revisited



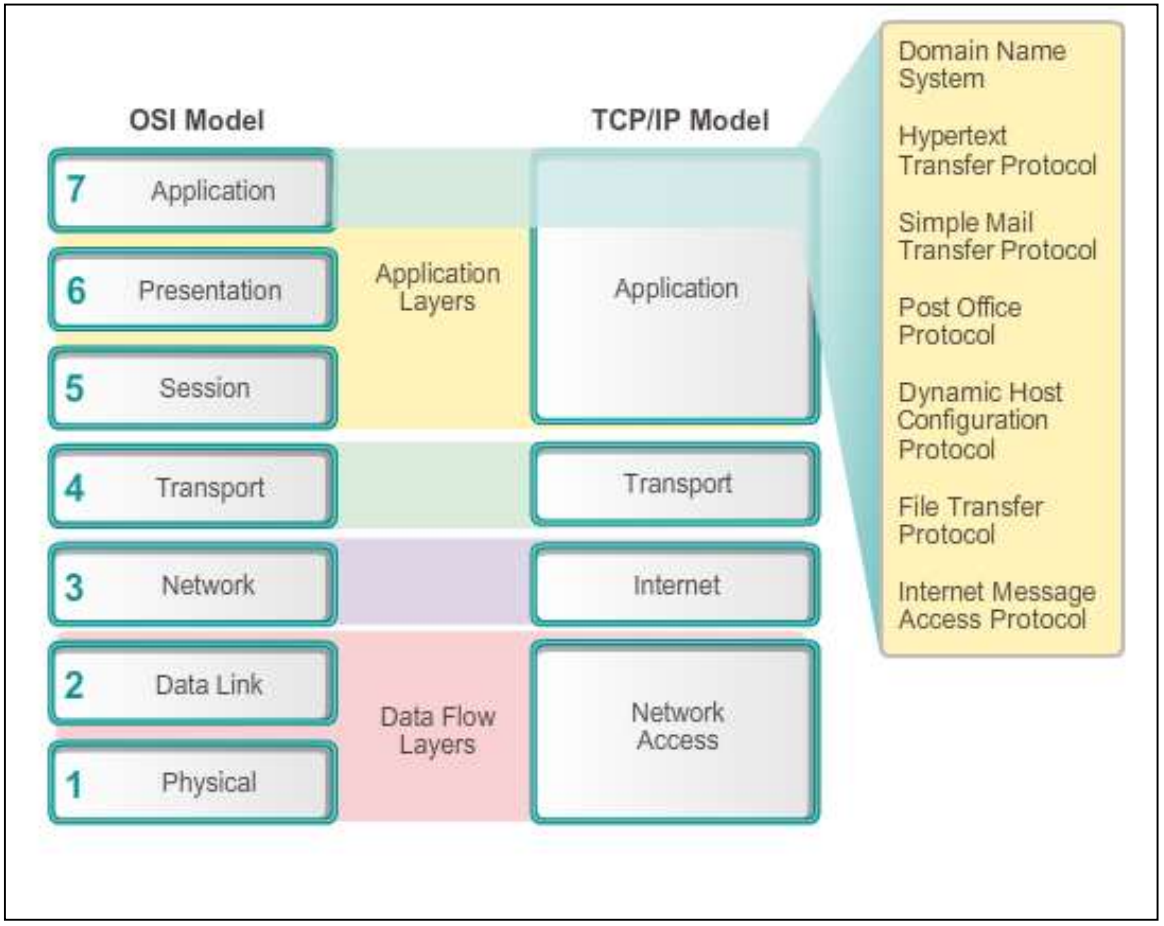
Provides the interface between the applications we use to communicate and the underlying network over which our messages are transmitted.

Application layer protocols are used to exchange data between programs running on the source and destination hosts.

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Application Layer

Most applications, like web browsers (using HTTP) or e-mail clients (using SMTP and PoP), incorporate the functionality of the application, presentation and session layers.



DNS
HTTP
DHCP
FTP

<u>Transfer</u>
SMTP

<u>Retrieval</u>
IMAP
POP

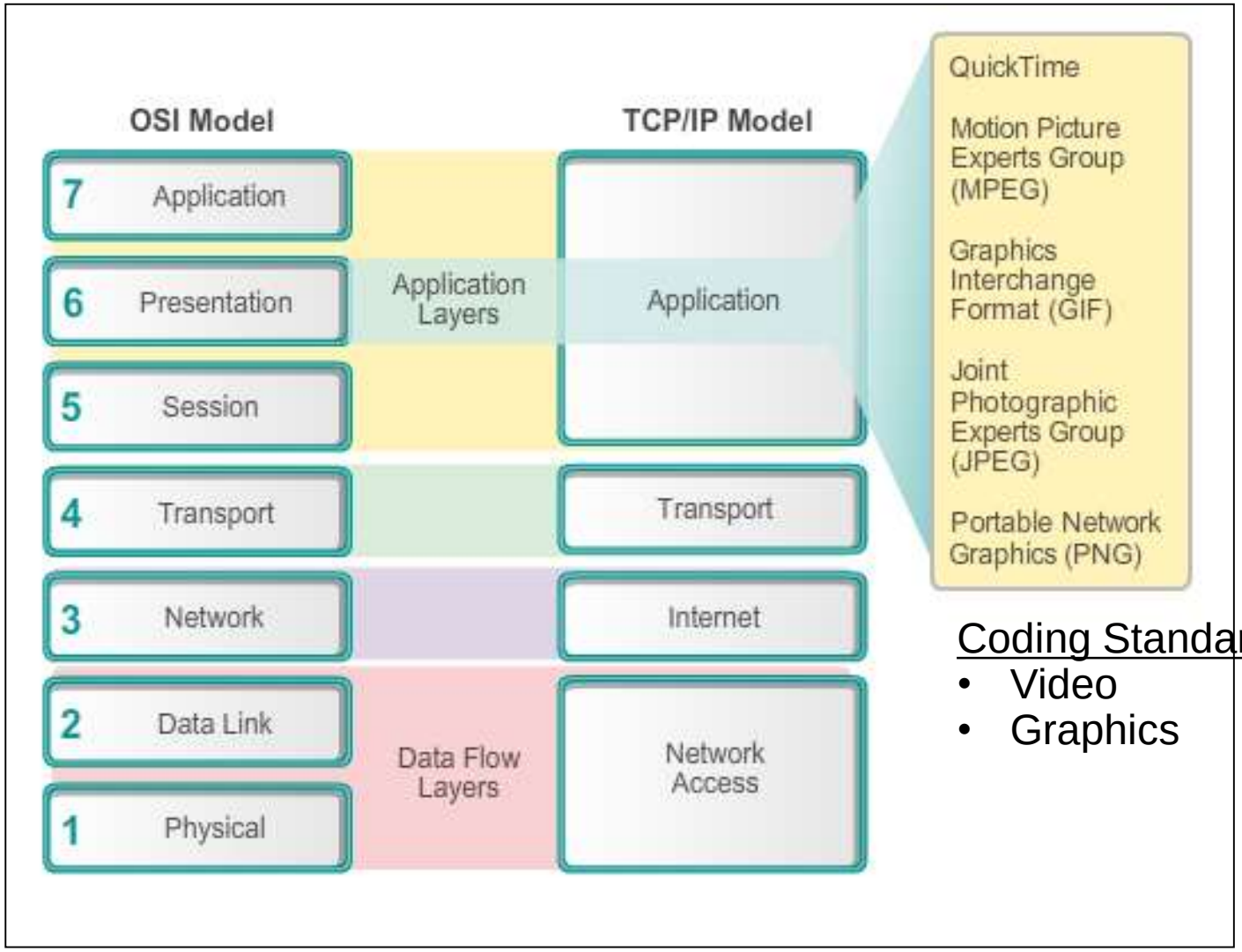
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Presentation and Session Layers

- **Presentation layer**
  - Coding and conversion of application layer data
  - Data compression
  - Data encryption for the transmission and decryption of data upon receipt by the destination
- **Session layer**
  - Functions, creates, and maintains dialogs between source and destination applications
  - Handles the exchange of information to initiate dialogs, keep them active, and to restart sessions

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Presentation and Session Layers (cont.)

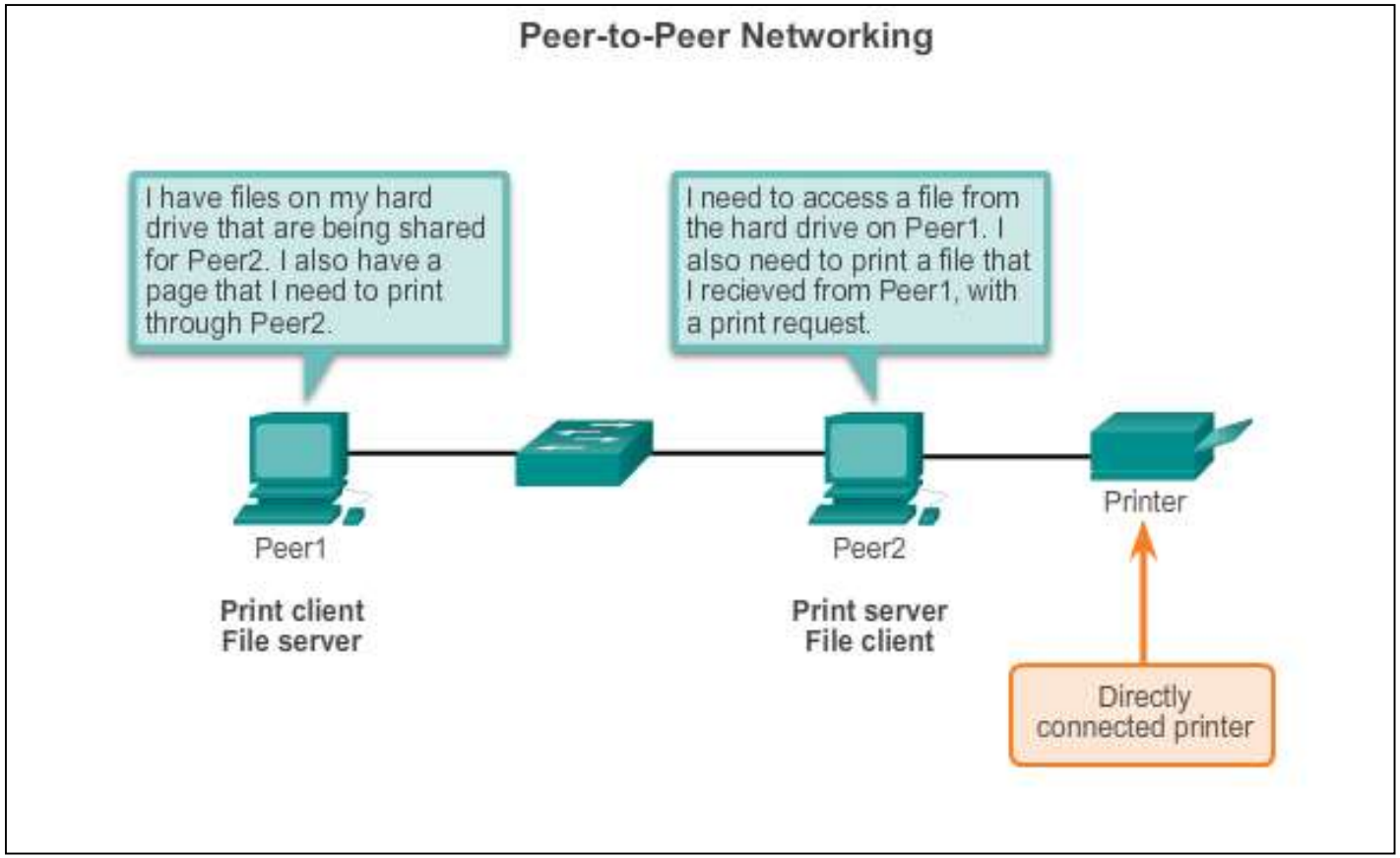


Coding Standards

- Video
- Graphics

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Peer-to-Peer Networks

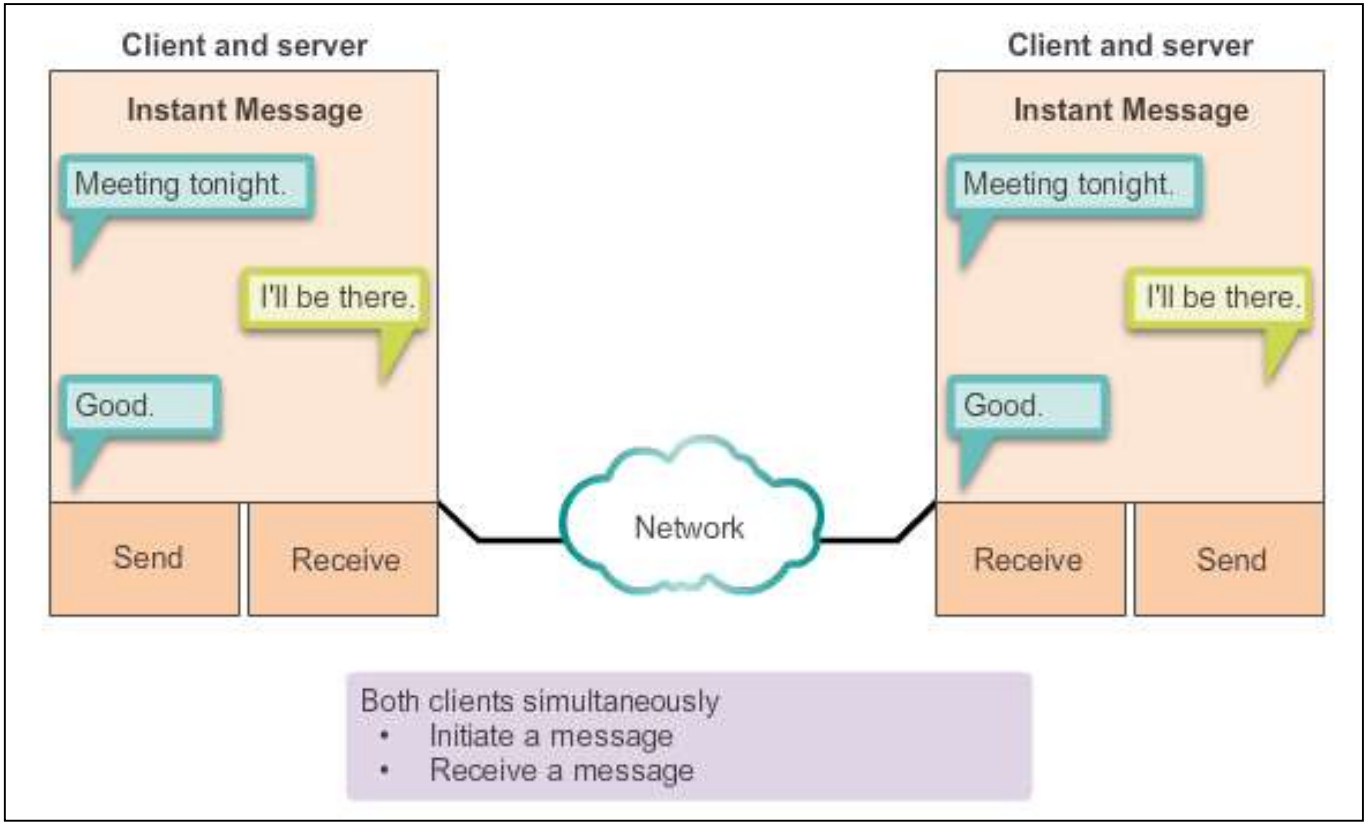


Both devices are considered equal in the communication. The roles of client and server are set on a per request basis.

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Peer-to-Peer Applications

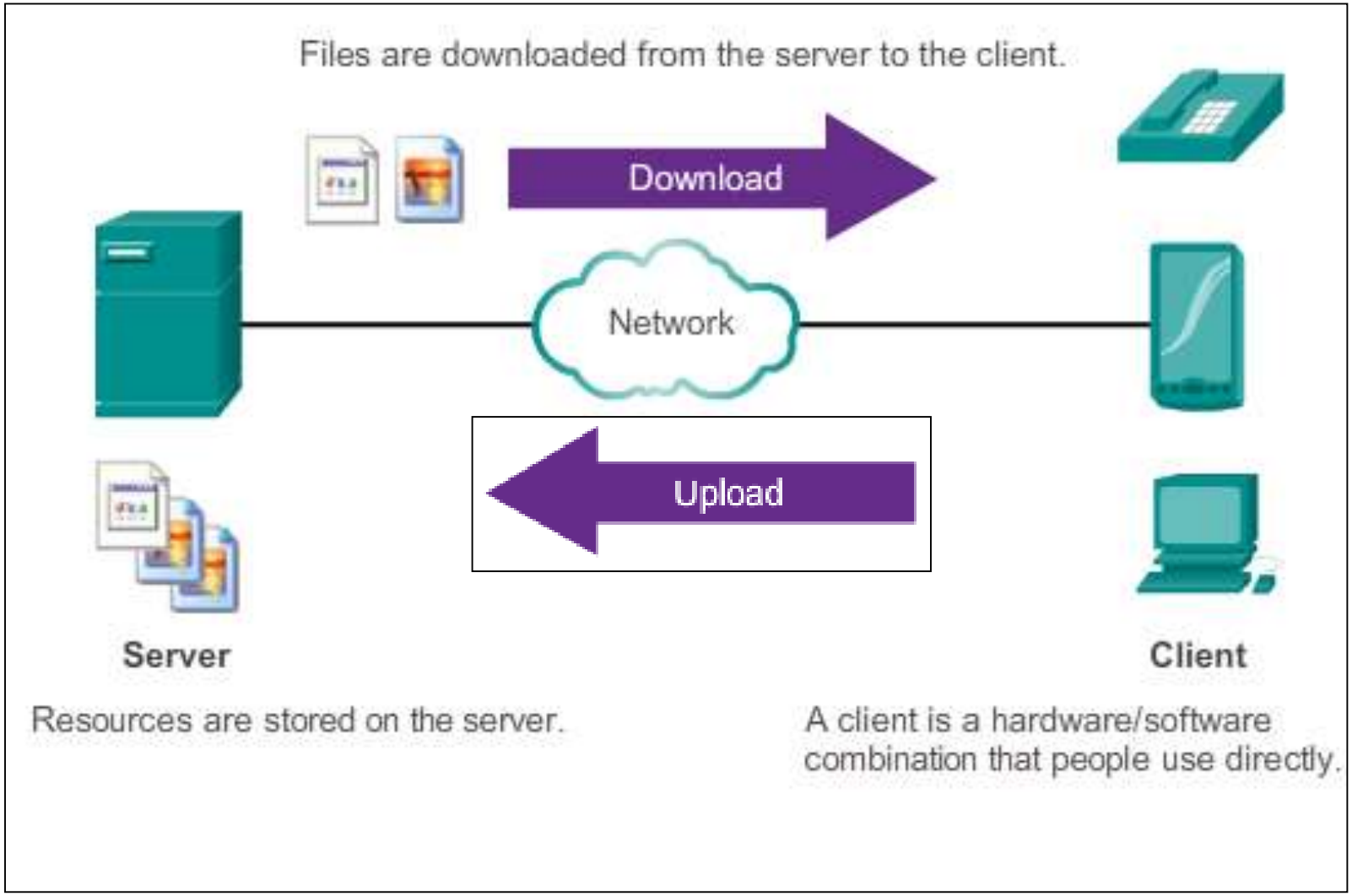
Client and server in the same communication.



Both act as client and server at the same time

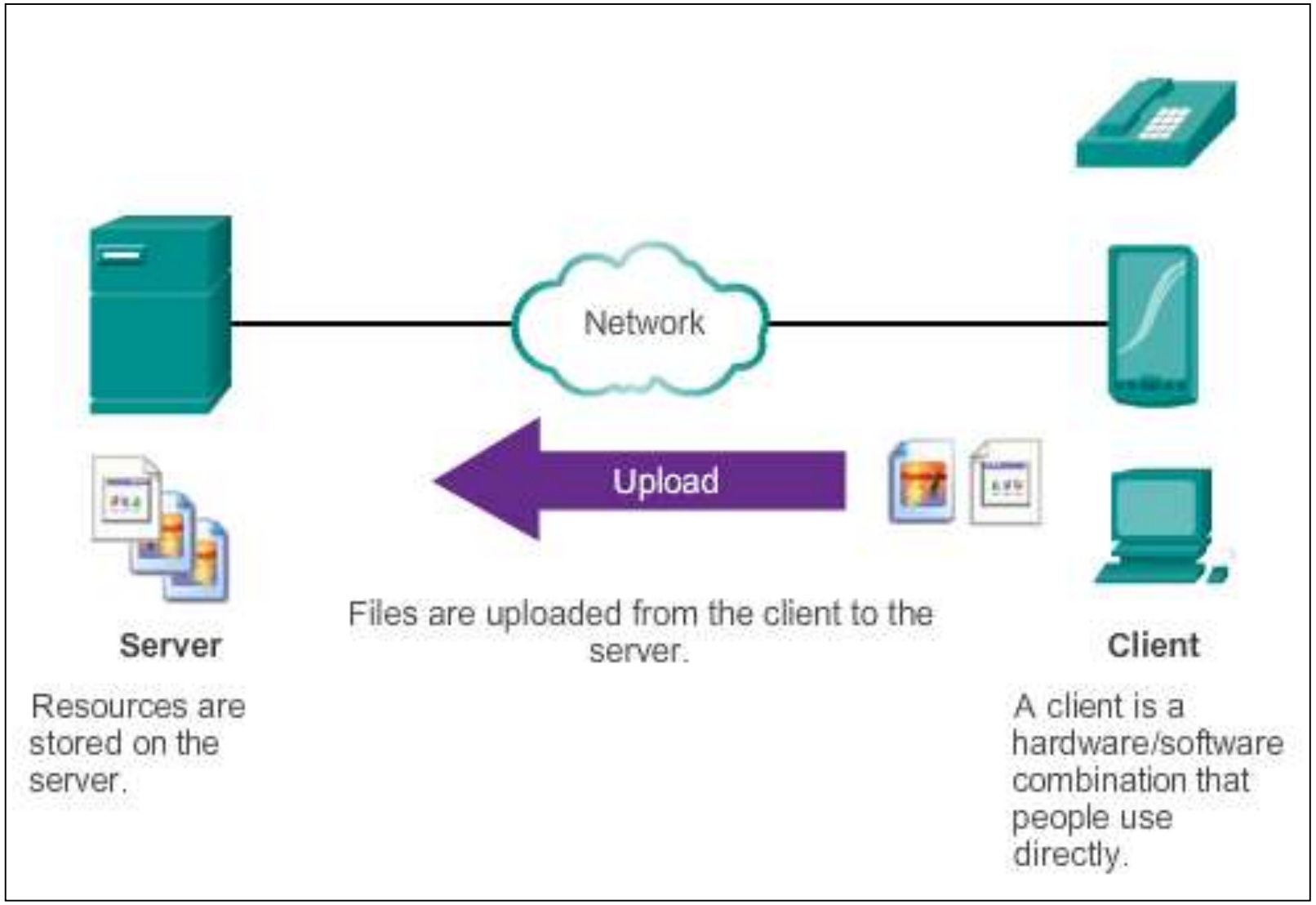
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Client-Server Model



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Client-Server Model



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10.2 Well-Known Application Layer Protocols and Services



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Application Layer Protocols Revisited

Three application layer protocols involved in everyday work or play include:

- **HTTP** to browse the web.
- **Simple Mail Transfer Protocol (SMTP)** to enable users to send email.
- **Post Office Protocol (POP**) to enable users to receive email.

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Hypertext Transfer Protocol and Hypertext Markup Language

Example URL: http://www.cisco.com/index.html

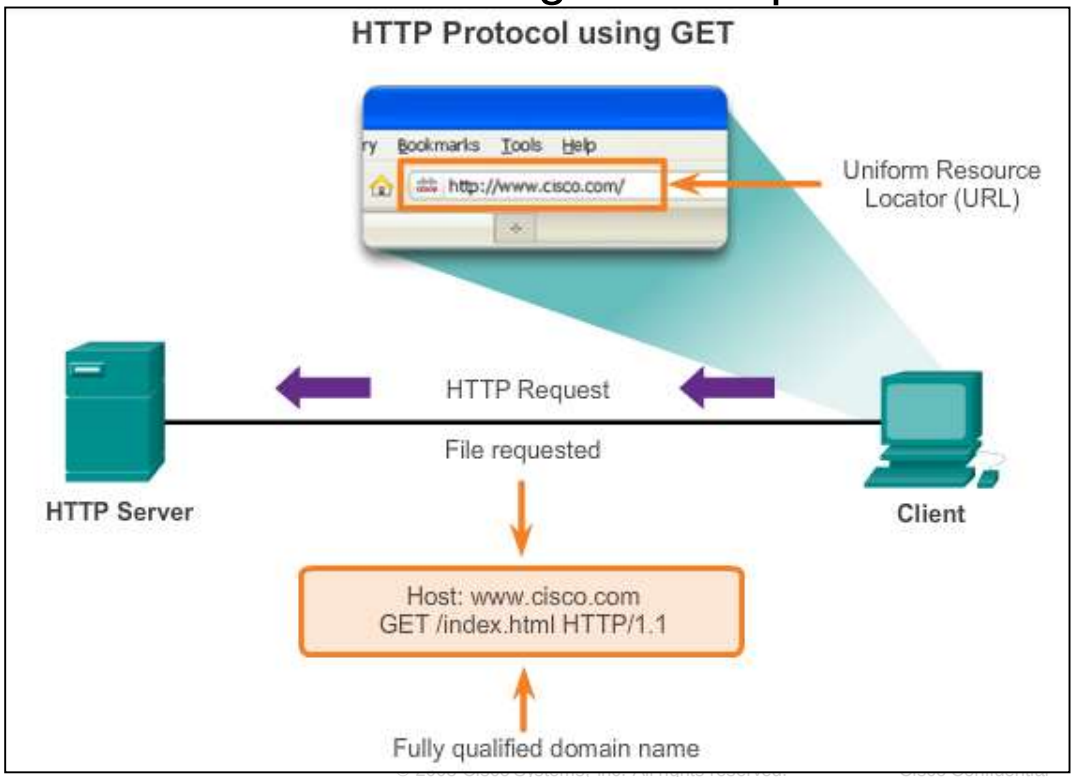
172.15.10.5

1. First, the browser interprets the three parts of the URL:
   - **http** (the protocol or scheme)
   - **www.cisco.com** (the server name)
   - **index.html** (the specific file name requested)
2. Browser checks with a name server to convert **www.cisco.com** into a numeric address
3. Using the HTTP protocol requirements sends a GET request to the server and asks for the file **index.html**
4. Server sends the HTML code for this web page
5. Browser deciphers the HTML code and formats the page

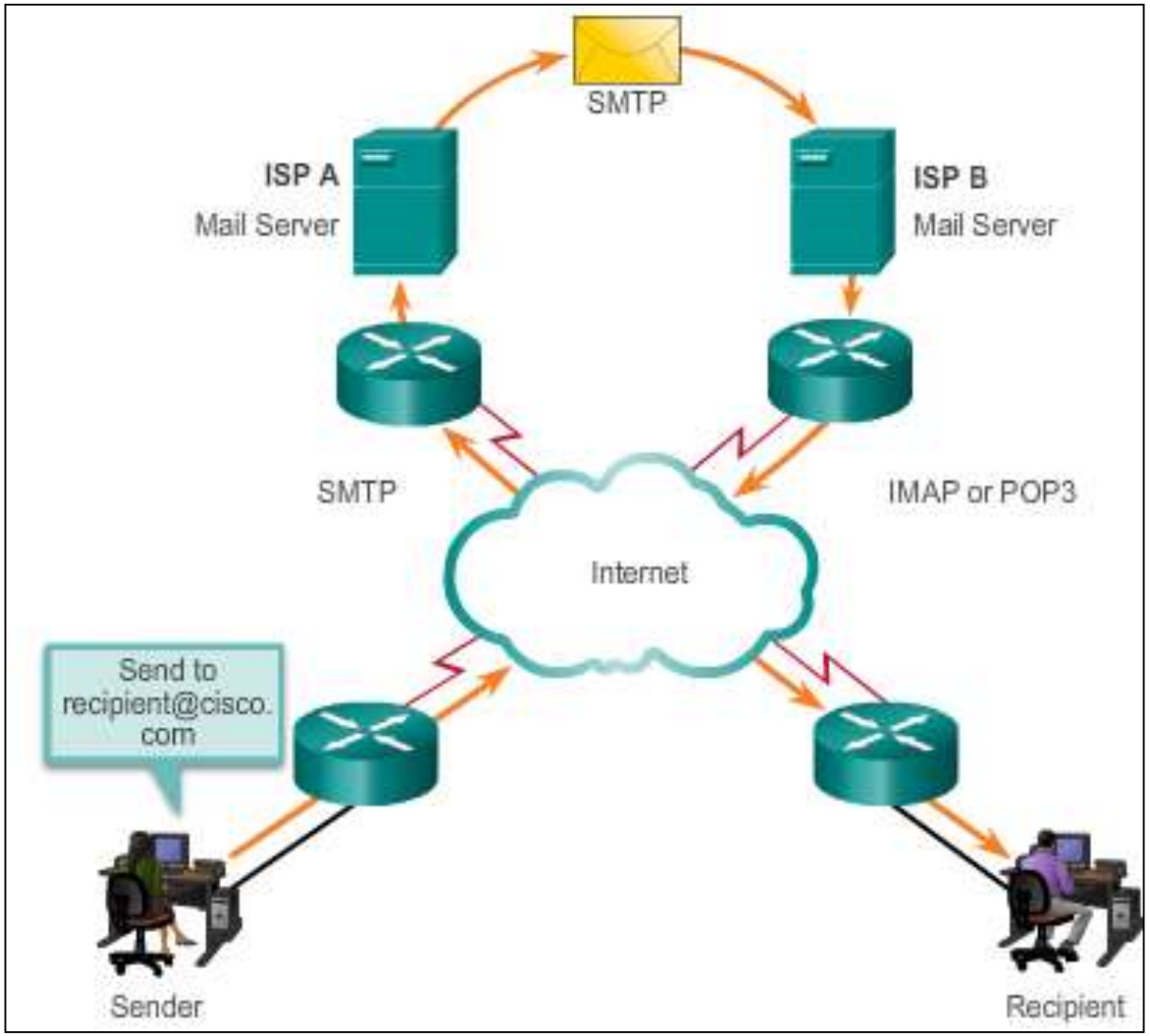
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HTTP and HTTPS

- Developed to publish and retrieve HTML pages
- Used for data transfer
- Specifies a request/response protocol
- Three common message types are **GET, POST**, and **PUT**
- GET is a client request for data
- POST and PUT are used to send messages that upload data to the web server



SMTP, POP, and IMAP (cont.)



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SMTP, POP, and IMAP (cont.)

Simple Mail Transfer Protocol (SMTP)

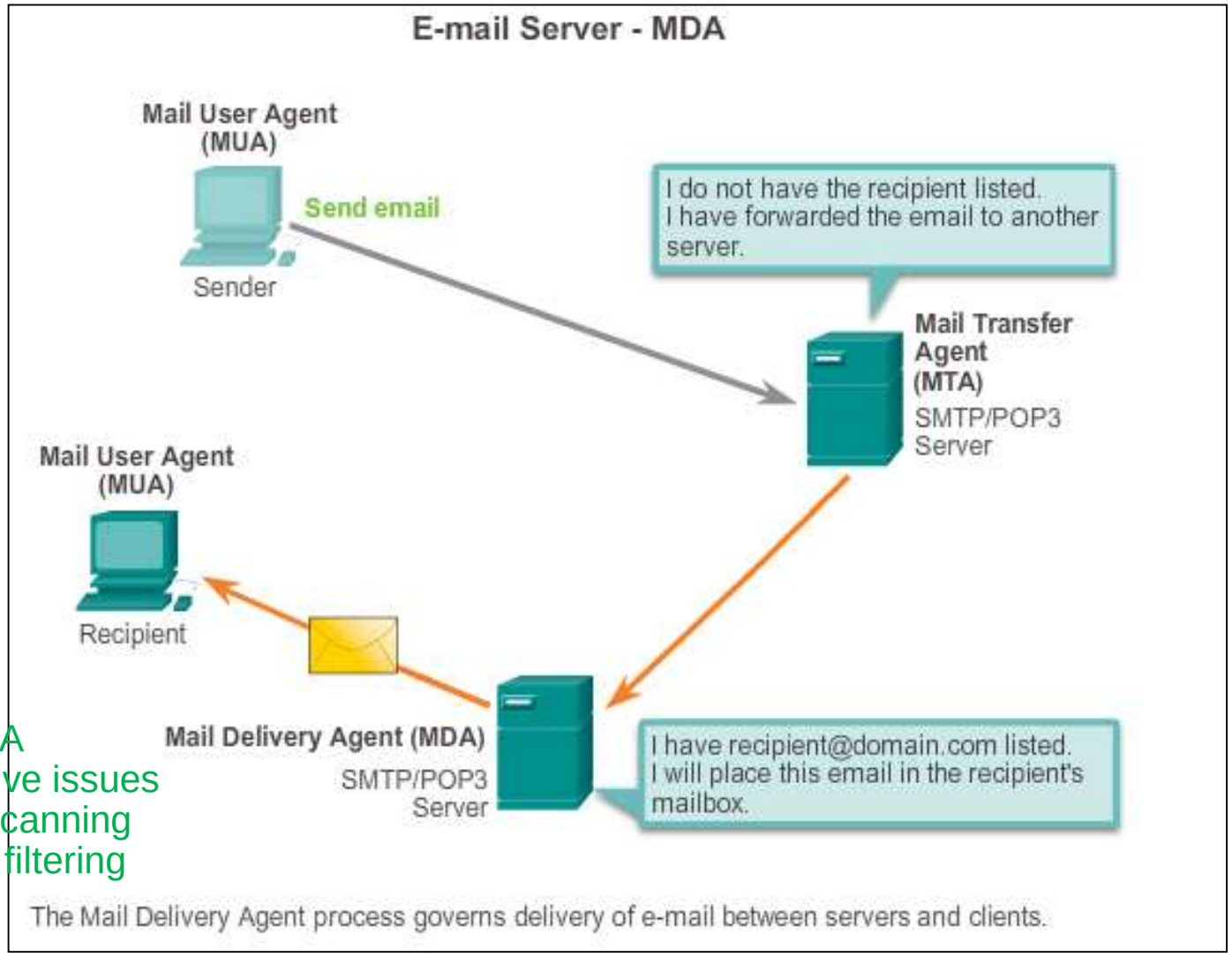
- **transfers mail**
- **SMTP processes must be running on both the client and server**
- uses **port 25**

Post Office Protocol (POP)

- enables a workstation to **retrieve mail** from a mail server
- mail is downloaded from the server to the client and then **deleted on the server (POP does not store messages)**
- uses **port 110**
- **POP3 is desirable for an ISP**, because it alleviates their responsibility for managing large amounts of storage for their email servers

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SMTP, POP, and IMAP (cont.)



MDA
Can also solve issues like virus scanning and spam filtering

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SMTP, POP, and IMAP (cont.)

Simple Mail Transfer Protocol (SMTP)
- Transfers mail reliably and efficiently

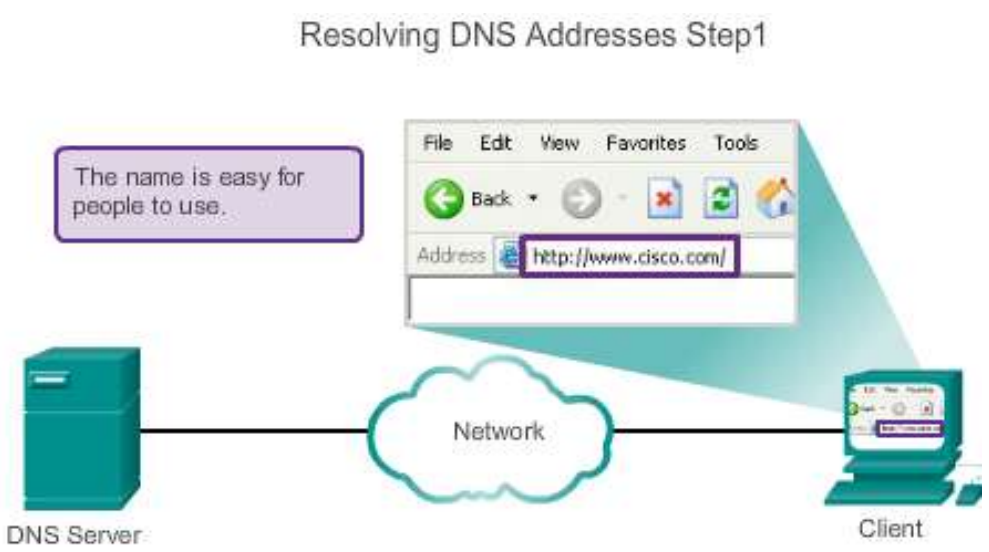
Post Office Protocol (POP)
- Enables a workstation to retrieve mail from a mail server
- With POP, mail is downloaded from the server to the client and then deleted on the server

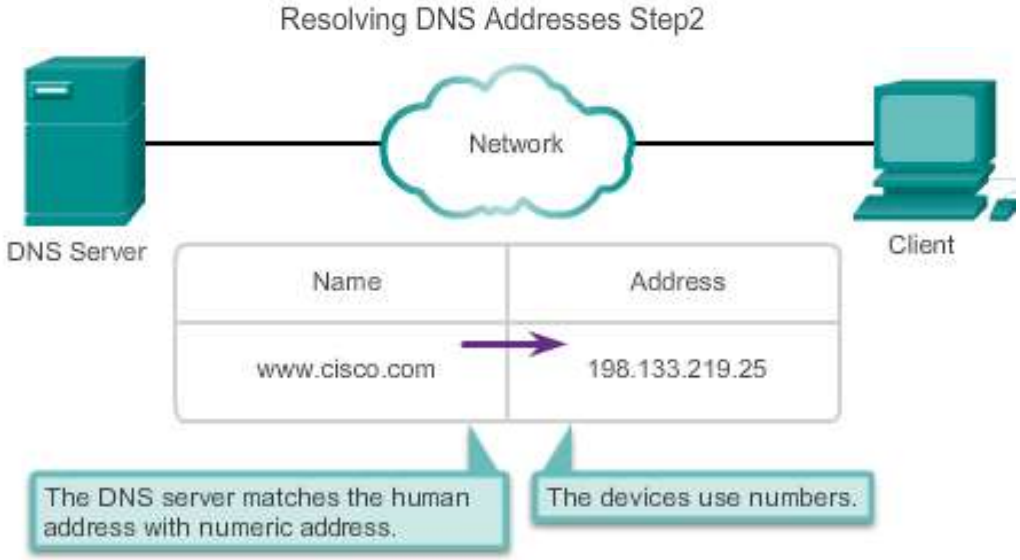
Internet Message Access Protocol (IMAP)
- Another protocol that retrieves email messages
- Unlike POP, when the user connects to an IMAP-capable server, copies of the messages are downloaded to the client application
- **Original messages are kept on the server until manually deleted**

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Domain Name Service

An URL name is resolved to its numeric network device address by the DNS protocol.





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DNS Hierarchy

Examples top-level domains (TLD):

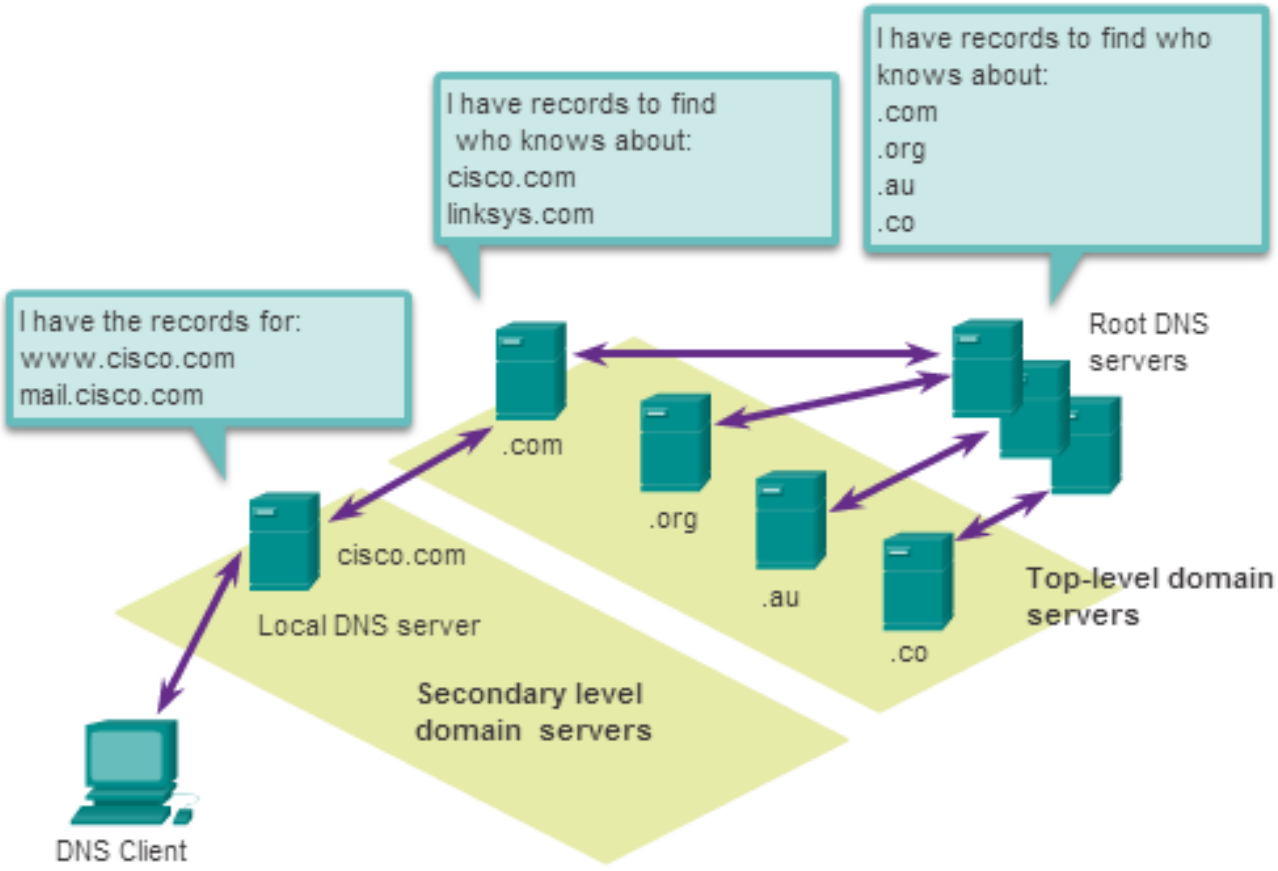
.au - Australia

.co - Colombia

.com - business or industry

.jp - Japan

.org - non-profit organization



A hierarchy of DNS servers contains the resource records that match names with addresses.

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nslookup

- Operating system utility called nslookup allows the user to manually query the name servers to resolve a given host name
- Utility can be used to troubleshoot name resolution issues and to verify the current status of the name servers

```
C:\Documents and Settings>nslookup
Default Server:  dns-sj.cisco.com
Address:  171.70.168.183

> www.cisco.com
Server:  dns-sj.cisco.com
Address:  171.70.168.183

Name:    www.cisco.com
Address:  198.133.219.25

> cisco.netacad.net
Server:  dns-sj.cisco.com
Address:  171.70.168.183

Non-authoritative answer:
Name:    cisco.netacad.net
Address:  128.107.229.50

>
```

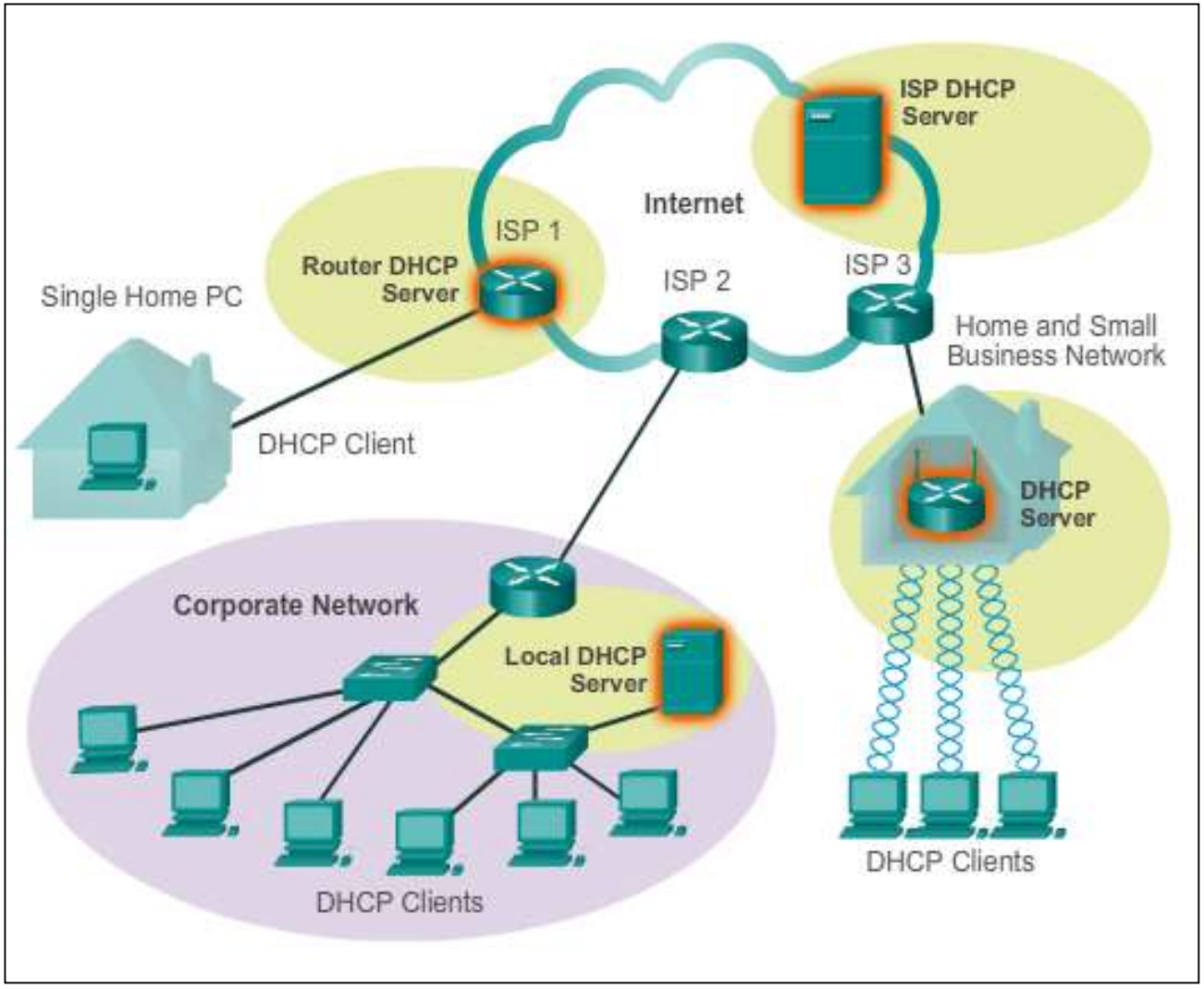

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Dynamic Host Configuration Protocol

- DHCP allows a host to obtain an IP address dynamically.
- DHCP server is contacted and address requested - chooses address from a configured range of addresses called a pool and “leases” it to the host for a set period.
- **DHCP used for general purpose hosts such as end user devices, and static addressing is used for network devices such as gateways, switches, servers and printers.**

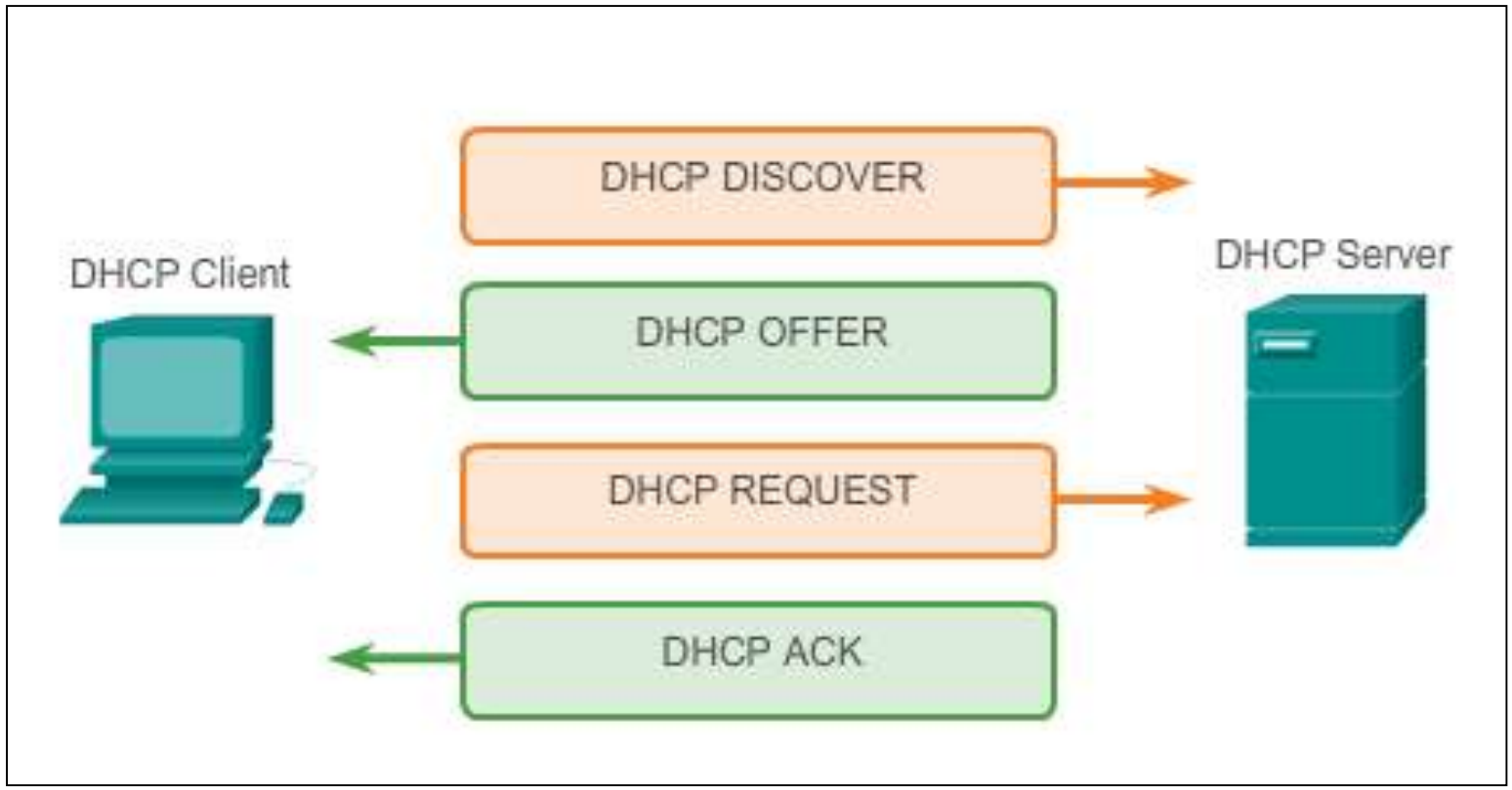
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Dynamic Host Configuration Protocol (cont.)



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DHCP Operation



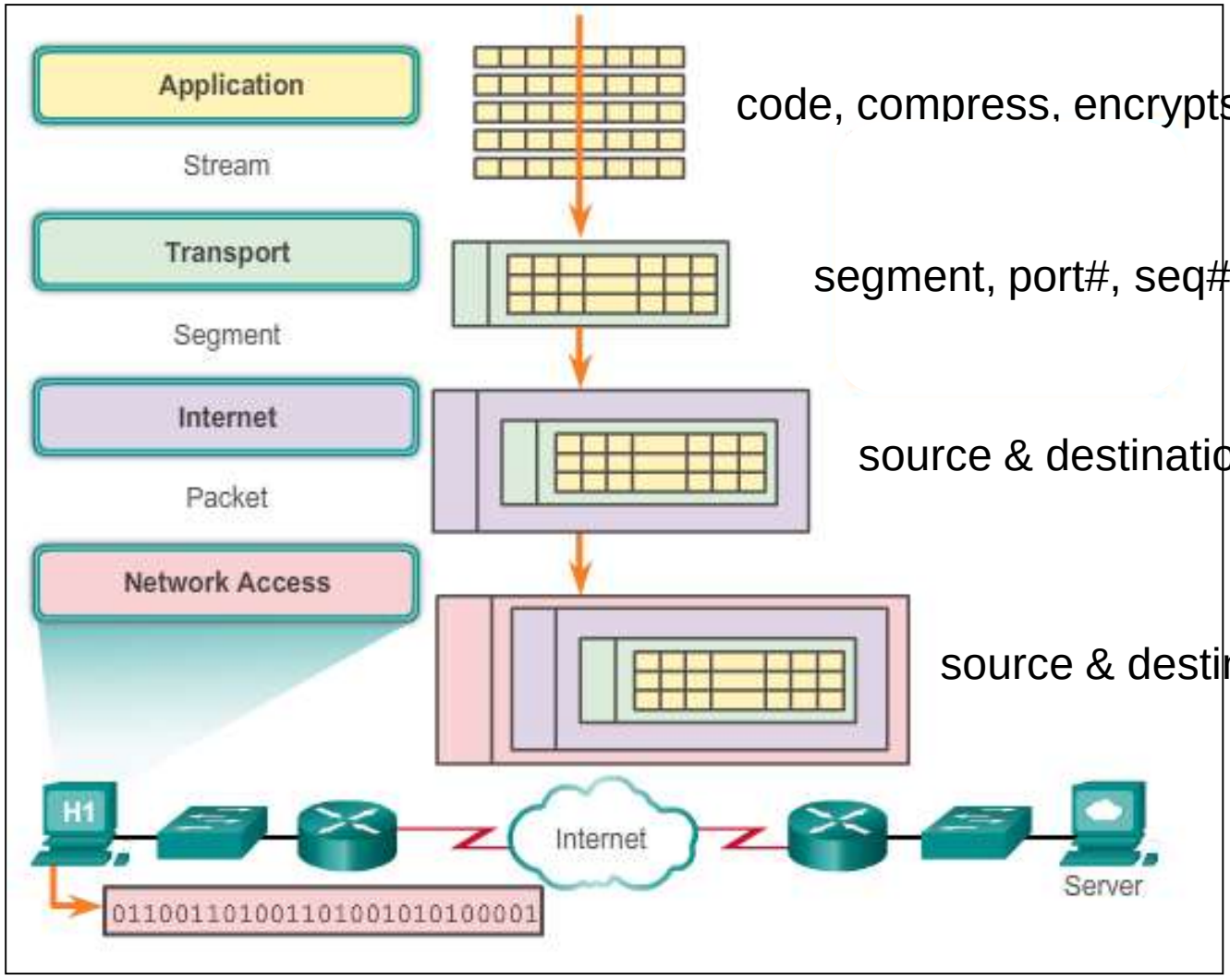
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File Transfer Protocol

- FTP allow **data transfers between a client and a server**.
- **FTP client is an application that runs on a computer that is used to push and pull data from a server running an FTP daemon.**
- To successfully transfer data, FTP requires two connections between the client and the server, one for commands and replies, the other for the actual file transfer.

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Getting the Data through the Internetwork



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Summary

- Applications are computer programs with which the user interacts and which initiate the data transfer process at the user's request.
- Services are background programs that provide the connection between the application layer and the lower layers of the networking model.
- Protocols provide a structure of agreed-upon rules and processes that ensure services running on one particular device can send and receive data from a range of different network devices.
- HTTP supports the delivery of web pages to end devices.
- SMTP, POP, and IMAP support sending and receiving email.

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